Investment Strategy Report

AI Infrastructure, ESG Review,
and Portfolio Performance

Team Name	**SWARM**
Team Members	Shawn, Reed, Willian
Main Stocks Covered	Marvell Technology (MRVL); Advanced Micro Devices (AMD)
Trading Case Discussed	Tesla (TSLA) short position as a risk-control lesson
Date	June 2026

Report Division of Work

Member	Main Responsibility	Contribution to Final Report
Shawn	Portfolio strategy, performance review, risk control	Organized the trading record, explained the long-short approach, reviewed portfolio results, and checked final consistency.
Reed	AMD research	Prepared the AMD business, competition, ESG, and valuation discussion.
Willian	MRVL research and source checking	Prepared the MRVL investment thesis, ESG review, and target-price appendix.

Prepared for SIC 2026 Season 15 - Junior Division

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Part I. Portfolio Strategy Overview

1. Current Portfolio Performance

At the end of the SIC S15 trading period, Team SWARM's account value was $125,650.32, which is an increase from their initial capital of $100,000. This implies that the portfolio has generated approximately 25.65% profit for the cycle. The final market value was $136,878.55, and the open profit and loss was +$26,336.53. These numbers indicate that our principal idea was successful, but also that this portfolio was not a low-risk or highly diversified portfolio. It was a focused approach and focus is a two-way street.

Starting capital	$100,000
Estimated portfolio return	25.65%
Open P&L	+$26,336.53

We had a picture: the biggest growth story in the stock market soon was going to be AI infrastructure. We focused on the hardware behind it not the consumer apps. We looked at chips, custom silicon, data-center networking and fast connections. This view helped us invest more in MRVL and AMD. At the time it made our portfolio sensitive to sudden changes in market sentiment, toward semiconductors. Looking back the idea was good. We should have had stricter rules for sizing our positions from the start.

2. Investment Strategy and Risk-Reward Profile

Our approach is a focused long-short equity approach. Companies that directly benefit from investments in AI infrastructure were on the long side, including Advanced Micro Devices (AMD) and Marvell Technology (MRVL). The short side was more carefully utilized and only in situations, where we felt a stock had become overvalued or had an imminent negative catalyst. The most obvious example of this aspect of the strategy was the TSLA short position, which failed.

The risk/reward profile was high. We took some margin during the trading period which enhanced both the potential reward and the potential loss. That meant three important factors for us: selecting a theme that was right, holding onto our positions when the thesis still held, and moving out of the positions as soon as the market proved us wrong. That's the reason the TSLA short was a valuable lesson. If the price goes against the position and continues to move, being on the logical side of the risk doesn't do much good. Time and momentum are as important as the thesis in short selling.

Again, if we had another chance we would continue our theme of AI infrastructure, but also spend time on our entry reason, trigger, and our max loss per position before we get in the trade. In the trading period some of these rules were brought up and discussed by the team, but not always laid out clearly. Having a written trading plan would have eliminated the hesitation and consistent risk taking.

3. Sector Focus and Trading Logic

We targeted semiconductor and data infrastructure because AI requires more than just a software story. All of this is infrastructure, from GPUs and CPUs, custom chips, optical connections, networking chips, memory bandwidth to advanced foundry capacity, that's required for large language models and cloud AI services. This compelled us to focus more on the companies that supply the "tools" of AI, as opposed to companies that tout about "AI" as a marketing term.

At MRVL, our entry logic was along custom silicon and electro-optics. At the time, we felt the market was starting to catch on to the fact that AI data centers require more than just GPUs. The data flow between servers and accelerators needs to be swift and efficient, which drives the need for MRVL products. We bought MRVL at $131.52, and valued the stock at $196.33 in our draft record for a 49.28% return.

For AMD we went with our entry logic that it had a data center CPU business and had an opportunity to expand in the AI accelerators. AMD does not have to be as successful as NVIDIA to be a successful investment. Its ability to earn more keeps growing even if it does only secure a viable share of the AI accelerator market, if it can continue increasing server CPU sales. It's for this reason that we considered AMD as a core long position, albeit with the understanding that it had already been priced in based on high expectations.

4. Position-Level Trading Review and Risk Control

Our trades were divided into three categories: core long trades, supporting long trades, and tactical short trades. Stocks that were most closely linked to our core thesis were given bigger position sizes. Supporting long positions gave us related exposure, but with smaller position sizes. Tactical shorts were used either to hedge the portfolio or to take advantage of short-term overvaluation, but they carried a different kind of risk.

The TSLA short was a useful mistake. We started shorting Tesla because we felt the interest in EVs would weaken and there would be pressure on the valuation. Instead, the price moved against us. We exited the position rather than waiting for the market to turn back in our favor. This decision helped preserve capital and limited the damage to the AI infrastructure longs. The lesson is easy to forget but very true: a short position may not have a clear floor on the loss, so discipline has to overcome pride.

Position Type	Purpose	Risk Control Rule
Core longs	Capture the AI infrastructure theme	Hold while the thesis and price action both remain strong
Supporting longs	Add related exposure without overloading one stock	Keep position size smaller and review catalysts regularly
Tactical shorts	Exploit overvaluation or negative catalysts	Exit quickly if momentum rejects the thesis

5. ESG Screening Framework

We did not include ESG as an afterthought at the end of this report. Instead, we connected it to the theme of AI + ESG investing. We used ESG as a second filter after the investment thesis. A stock needed to benefit from AI growth, but it also had to show that its growth could be responsible and sustainable. Our most relevant ESG questions were about energy use, supply-chain responsibility, data and AI ethics, board oversight, and whether management incentives are linked with long-term value.

Dimension	What We Checked	MRVL View	AMD View

Environmental	Emission goals, data-center energy use, product efficiency, supplier footprint	Positive on product efficiency and reported Scope 1/2 progress; still exposed to Scope 3 and foundry partners.	Positive on 2030 Scope 1/2 reduction target and performance-per-watt; still dependent on suppliers such as TSMC.
Social	Supply-chain labor standards, responsible sourcing, privacy, AI ethics	Mixed-positive. Its products support key AI infrastructure, but supply-chain transparency must keep improving.	Mixed-positive. AMD enables broad AI adoption, so it must manage privacy, cybersecurity, and supplier responsibility carefully.
Governance	Board independence, executive pay, shareholder alignment, ESG oversight	Generally positive: independent board structure and performance-linked compensation support our view.	Generally positive: mostly independent board, ESG oversight, stock-based incentives, and clawback policies.
Investment use	How ESG changed our decision	Supported a Buy view, but made us watch power use and customer concentration more closely.	Supported a Buy view, but did not remove competition, supplier, and valuation risks.

This ESG review affected our ranking. Our top idea was still MRVL because its AI networking exposure is directly tied to data-center efficiency. AMD was second because the market opportunity is larger, but the competition and valuation risks are also higher. In both cases, ESG did not replace financial analysis. It helped us ask whether the growth story could last.

Part II. Best Investment Ideas

1. Marvell Technology, Inc. (MRVL)

Marvell Technology was one of the strongest investment ideas in this report. We chose Marvell Technology because it is directly involved in AI data-center infrastructure, especially custom silicon, networking chips, and electro-optical connectivity.

In our view, investors first paid the most attention to graphics processing units and the most well-known artificial intelligence companies. However, the next stage of AI growth also depends on companies like Marvell Technology, because data centers need faster and more efficient ways to move data between chips and servers.

For this reason, we viewed Marvell Technology as a long-term investment opportunity rather than just a short-term trade.

Company Overview and Investment Thesis

Marvell Technology makes computer chips that help store and send data. These chips are used in computer systems. We thought buying Marvell Technology stock was an idea. It made sense to us. We bought the stock for $131.52. Then it went up to $196.33. That is a gain of 49.28%.

The main thing to remember is that big computer systems need a lot of things to work well. They need more than fast processors. They also need chips that can move data quickly connections and optical cables. Marvell Technology is a company that makes these things. Marvell Technology may not be well known as some other companies but it is very important. Marvell Technology is not, like NVIDIA it is different. Marvell Technology can still benefit from people using intelligence more and more. We saw Marvell Technology as a company that would do well once people started looking at companies besides the obvious ones that work with artificial intelligence. Marvell Technology is part of the intelligence industry that people do not always notice. Marvell Technology is very important.

Industry and Competitive Analysis

Porters Five Forces make it clear that MRVL is a choice but it is not completely safe. It is hard for new companies to enter this field because customers want suppliers who have a track record can design things well and can get the job done. The customers, the big cloud companies have a lot of power because they buy a lot but they also need stable Artificial Intelligence infrastructure so they cannot push the suppliers too much. The suppliers have a to high amount of power because it is hard to find companies that can make advanced parts package them and make optical components.

The competition is very strong. What matters most is the quality of MRVLs engineering and their relationships with customers not just the price. The threat of companies making their own products is moderate. Big cloud companies might start designing their chips, which could mean they need outside vendors like MRVL less. However even if they design their chips they will still need help with networking, connectivity and getting everything to work together. Switching to a supplier is also expensive and risky. So there is a risk that MRVL could lose customers to other companies but it is not big enough to change the idea that MRVL is a good choice, for the next year.

Financial Overview and Valuation

We didn't treat MRVL as a traditional low P/E value stock as it trades at a premium valuation. We chose a growth-oriented relative valuation model in Appendix A to arrive at our $220 target price. The base case scenario assumes that AI custom silicon and electro-optics continue to expand at a higher rate than the rest of the company and that operating leverage will improve as the higher growth segment of the company expands.

Multiple contraction is the main downside case. Even if the business performs well, MRVL's stock could fall if cloud capital expenditure slows or if investors decide that AI infrastructure stocks are overvalued. That is why we used a sensitivity table instead of treating one target price as a guaranteed number. It is a prediction, not a prophecy.

ESG Review: MRVL

Environmental: MRVL is not a “green” company simply by selling chips for AI. The environmental issue is whether data centers can help customers get more data through with less power—MRVL does just that. This is why the performance per watt is important in the context of ESG and business value. Marvell has announced a science-based target of 50% reduction in Scope 1 and Scope 2 greenhouse gas emissions by FY2030 from the FY2022 base-year and has made disclosures to demonstrate progress towards that goal. The other weakness is Scope 3, as a lot of semiconductor footprint is in the supply chain.

Social: MRVL's social risks are related to labor standards in its supply chain, responsible sourcing, cybersecurity, and the social footprint of AI infrastructure. We rated this area as mixed-positive. The company supports infrastructure that can improve productivity, but AI growth also creates concerns about privacy, job displacement, and access to technology. Supplier standards and responsible sourcing disclosures would continue to matter for our investment decision.

Governance: We found MRVL's governance to be enabling our investment case. The proxy disclosure states that its board is mostly independent, has a lead independent director, executive compensation is tied to performance, and it has stock ownership guidelines and a clawback policy. These features are not a guarantee of no risk but they do minimize the likelihood that management will be rewarded for short-term excitement alone. For us, the important takeaway here is that there is a governance "growth story" to which we can subscribe.

Risks and Catalysts

Catalysts: continued spending on AI data centers, increased orders for custom silicon, increased demand for electro-optics and forward guidance.

Risks include: Data-center energy policy focus, supply-chain constraints, customer concentration, cloud capital expenditures growth slowdown, and valuation compression.

2. Advanced Micro Devices, Inc. (AMD)

Advanced Micro Devices was one of our main investment picks in this report. We chose AMD because it gives our portfolio exposure to AI computing, data-center CPUs, and AI accelerators. Compared with Marvell Technology, AMD is more widely known, and its stock price already includes high expectations from the market. However, we still viewed AMD as a long-term investment opportunity because the need for AI computing power continues to increase. Many cloud customers are also looking for more options. They do not want to rely on only one major AI chip supplier, especially NVIDIA. This creates room for AMD to grow, even if it does not become the largest player in the AI accelerator market.

Company Overview and Investment Thesis

AMD designs CPUs, GPUs, adaptive computing products, and data-center accelerators. AMD is one of the few semiconductor companies with the scale, engineering expertise, and customer relationships needed to compete in both servers and AI accelerators. In our portfolio, AMD was not the top idea, but it was a core long position that gave us large-cap exposure to AI compute growth.

We do not need to prove that AMD is better than NVIDIA. The more realistic argument is that the AI accelerator market is large enough for AMD to build a meaningful second-place business while continuing to expand its server CPU market. If AMD can keep growing data-center revenue, it may improve margins and still deserve a premium valuation.

Industry and Competitive Analysis

AMD’s barriers to entry are extremely high. Designing semiconductors calls for high R&D investments, long product cycles, intellectual property, and high customer trust. Supplier power is also quite high since AMD relies heavily on advanced foundry and packaging partners, particularly at the leading edge nodes. The degree of customer power is moderate to high as hyperscale customers buy in large quantities while also seeking alternatives to not depending on a single supplier of AI hardware.

There is a risk that customers might choose something like custom chips made by cloud companies of AMDs products. This is not a huge issue at the moment. If a customer wants to switch from AMD they have to make changes. They need to get software and tools test the new products and change their buying process. This gives AMD time to improve its products. The biggest challenge for AMD is that other companies, like NVIDIA and Intel are making products and are rivals. NVIDIA has a lot of software that works with its products. Intel is a player, in the CPU market. AMD and NVIDIA compete with each other in the AI hardware market. Both are trying to make better products. AMD needs to keep improving to stay of its competitors. NVIDIA and Intel are competitors and AMD has to work hard to beat them.

Financial Overview and Valuation

In appendix A, the relative valuation model is used to determine the target price for our stock, which is $500. The model incorporates an expected forward EPS and target forward P/E. This isn't intended to be a comprehensive DCF model. We do not have enough information to make the necessary assumptions to complete a full DCF. The relative model is easier to understand and is more straightforward: it indicates the required earnings level and multiple to make the target price sensible.

The positive side of the equation is built on continued expansion of the data-center CPU market as well as increased uptake of AMD's Instinct AI accelerators. The bad case, which is clear, is that the sales of AI accelerators are underwhelming or that customers stay in NVIDIA's ecosystem. Even if the company is a good one, it can be a bad short-term investment when the price reflects excessive optimism.

ESG Review: AMD

Environmental: AMD's ESG story starts with energy efficiency. The company has declared its ambition to lower the Scope 1 and Scope 2 GHG emissions from its operations by 50% by 2030 from 2020. Product energy efficiency is also a focus of AMD, as it is important because AI data centers require greater compute at a rate that doesn’t increase power bills. Performance per watt isn't a "sustainability term" for investors – it impacts investors' economics.

AMD's fabless model is the main environmental constraint. AMD relies on manufacturing partners, so much of its environmental impact is indirect. This does not mean AMD is responsible for everything its suppliers do, but it does mean supplier oversight is important. Because AMD continues to monitor supplier emissions and manufacturing risk, it passed the environmental filter in our ESG screen.

Social: Supplier labor standards, responsible sourcing, AI ethics, privacy, cybersecurity, and the impact of AI on employment are social risks facing AMD. We do not believe a chip company can ignore these issues by saying it only provides hardware. The hardware makes AI systems used by companies and governments possible, so responsible product use and disclosure remain relevant. On the social dimension, we considered AMD investable but not risk-free.

Governance: This is not just a governance review of “Lisa Su is a strong CEO.” While leadership is important, so is the structure of the board and incentives for good governance. In proxy materials, AMD states that its board is mostly independent, that it has independent committees, that it has ESG oversight at the board level, that it has performance-based executive compensation, that it has stock ownership guidelines and that it has clawback rules. These characteristics are contributing factors to our Buy rating since they are based on a linkage of management incentives to long-term performance and not short-term market noise.

Risks and Catalysts

Catalysts: increased demand for Instinct accelerators, continued server CPU share gains, greater enterprise AI adoption, and greater demand by cloud customers for a second supplier.

The risks include the software edge NVIDIA holds, overvaluation, reliance on suppliers, geopolitical risk, and increased regulation on AI usage of power and data security.

Part III. Portfolio Performance and Attribution

The majority of our portfolio return was driven by longer term positions in AI infrastructure. MRVL was the most obvious winner as it had a thesis we could explain and defend, and was moving well on the price chart. AMD contributed to the large-cap AI compute exposure and the overall portfolio. TSLA was an underperforming stock on the short side, but the loss was contained with the change of price action.

Source of Return	Impact	Comment
MRVL long	Strong positive	Best individual idea and strongest realized contribution.
AMD long	Positive	Core AI compute exposure with continued upside but higher valuation risk.
TSLA short	Negative but controlled	Covered when price action rejected our short thesis.
Use of margin	Amplified returns and risks	Helpful in a rising AI hardware market, but should be used with stricter limits.

We still believe AI infrastructure can remain strong over the next six months, but we also think the easiest gains may already have been made. We would be more selective and reduce leverage if semiconductor valuations remain high. Companies most closely connected to AI data-center capital expenditure are still our preferred picks for the next 12 months. Over a three-year horizon, we would expand our research into edge AI, enterprise AI software, and energy-efficient infrastructure.

Our main improvement would not be giving up the AI thesis. It would be improving the process. We would not enter a trade without first writing down the thesis, the exit signal, and the maximum acceptable loss. We would also use a formal ESG checklist. The aim is not to find perfect companies. Perfect companies usually exist only in marketing slides. The goal is to avoid companies whose growth depends on risks that management refuses to acknowledge.

Conclusion

We believe that our team might not have done a job in SIC15 but we all tried our best. The AI infrastructure theme did well. Gave us a positive return and our main ideas, MRVL and AMD worked out as we thought they would. MRVL was the investment idea because it gave us a way to invest in custom silicon, data-center networking and electro-optics directly. AMD was our core idea because it was a leader in the area of data-center CPUs and AI accelerators.

Our portfolio was not perfect though. We used borrowed money to invest. Our decision to short sell TSLA showed that even a good idea can fail if the timing and price are not right. In the version of our strategy we will make some changes. We will be more careful about how much we invest in each company we will make a plan before we buy or sell. We will pay more attention to how companies treat the environment and their employees. We will look for AI companies that are making progress in using energy being responsible, with their supply chain and making sure their leaders are doing what their shareholders want.

Footnotes and References

1. Marvell Technology, FY2025 Sustainability Report: discussion of Scope 1 and Scope 2 greenhouse gas reduction targets and sustainability progress.
2. Marvell Technology, responsible sourcing and human rights materials: discussion of conflict minerals, supply-chain expectations, and supplier responsibility.
3. Marvell Technology, 2025 Proxy Statement: discussion of board independence, lead independent director, executive compensation, stock ownership guidelines, and clawback policy.
4. Advanced Micro Devices, 2024-2025 Corporate Responsibility Report: discussion of 2030 Scope 1 and Scope 2 greenhouse gas reduction target and product energy-efficiency priorities.
5. Advanced Micro Devices, 2025 Proxy Statement: discussion of ESG oversight, board independence, committee structure, executive compensation, equity incentives, stock ownership guidelines, and clawback policy.
6. Team SWARM SIC trading account records and draft report data: ending account value $125,650.32, market value $136,878.55, open P&L +$26,336.53, MRVL purchase price $131.52 and value $196.33, AMD and TSLA trade notes.

Appendix A. MRVL & AMD Valuation Model

We used a simple relative valuation model because a full DCF model would require many assumptions that are difficult to prove in this report. This model is not perfect, but it shows how our target prices were calculated instead of simply guessed.

The formula we used is: Expected Forward EPS x Target Forward P/E = Implied Target Price.

Company	Role in Portfolio	Expected Forward EPS	Target Forward P/E	Implied Price	Report Target Price
AMD	Core long position	$7.14	70.0x	$499.80	$500
MRVL	Best investment idea	$3.38	65.0x	$219.70	$220

For AMD we used an expected earnings per share of $7.14 and a target price-to-earnings ratio of 70.0 times. This gives us an implied stock price of $499.80, which we rounded to $500. We gave AMD a multiple because AMD has strong exposure to artificial intelligence computing. AMD also has a role in data-center CPUs and AI accelerators.. This target also depends on AMD continuing to grow in the AI accelerator market. It also depends on AMD keeping investor confidence.

For MRVL we used an expected earnings per share of $3.38 and a target price-to-earnings ratio of 65.0 times. This gives us an implied stock price of $219.70, which we rounded to $220. We gave MRVL a multiple than AMD because MRVLs growth depends more on custom silicon. MRVLs growth also depends on AI networking execution. We still think MRVL is our investment idea. This is because of MRVLs connection to AI data-center infrastructure.

The main weakness of this model is contraction. If the market stops giving high valuation multiples to AI infrastructure stocks both target prices, for AMD and MRVL would become less realistic. This is even if AMD and MRVL continue to grow. Therefore the target prices should be understood as base-case estimates. They are not guaranteed outcomes.

Sensitivity Check

Company	Bear P/E	Base P/E	Bull P/E	Bear Price	Base Price	Bull Price
AMD	60.0x	70.0x	80.0x	$428.40	$499.80	$571.20
MRVL	55.0x	65.0x	75.0x	$185.90	$219.70	$253.50

Our target prices for these stocks are really dependent on the Price to Earnings multiple. For AMD and MRVL the reason we like these companies is still good, in the situation but if things go badly the price of these stocks could be a problem. This is why we think people should buy AMD and MRVL. We do not think they are completely safe investments.